SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

CHAPTER 07

How Do Organisms Reproduce?

(Full Chapter)

> DNA Copying & Importance of Variation • Modes of Asexual Reproduction • Sexual Reproduction in Flowering Plants • Human Reproductive Systems • Sexual Health & Contraception

Target Level: CBSE Class 10

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📖 Chapter Overview & Basic Concepts

Reproduction is the biological process by which existing organisms give birth to new individuals of the same species. While not essential for an individual's survival, it is vital for maintaining population continuity and stability.

- **DNA (Deoxyribonucleic Acid):** Found in chromosomes within the nucleus, DNA contains hereditary information passed from parent to offspring.
- **DNA Copying:** Cells build chemical copies of their DNA prior to cell division. Slight biochemical inaccuracies during copying create structural variations.
- **Importance of Variation:** Variations enable species to adapt to changing environmental niches (e.g., sudden temperature changes), preventing complete extinction.

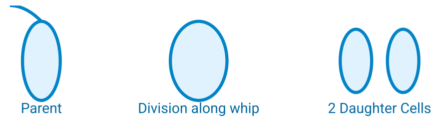
> **NCERT High-Yield Insight:** No biochemical process is 100% accurate. Minor inaccuracies during DNA replication lead to variations. Over generations, these beneficial variations accumulate and form the primary basis for biological evolution.

SECTION 1: MODES OF ASEXUAL REPRODUCTION

Asexual reproduction involves a single parent producing offspring without the formation or fusion of gametes. Offspring are genetically identical to the parent.

Mode	Mechanism	NCERT Examples
Fission	Splitting of a parent cell into two (Binary) or multiple (Multiple) daughter cells.	*Amoeba* (Binary), *Leishmania* (Binary along whip-like structure), *Plasmodium* (Multiple).
Fragmentation	Parent breaks into smaller pieces upon maturation; each fragment grows into a new organism.	*Spirogyra* (Multicellular filament).
Regeneration	Specialized cells proliferate to regrow complete body parts from cut body fragments.	*Planaria*, *Hydra*.
Budding	A bulb-like outgrowth (bud) develops due to repeated cell division at a specific site, detaching when mature.	*Hydra*, Yeast.
Vegetative Propagation	New plants develop from vegetative parts (root, stem, leaf) instead of seeds.	*Bryophyllum* (leaf notches), Potato (tuber eyes), Layering/Grafting (Rose, Sugarcane).
Spore Formation	Spores formed in bulbous structures (Sporangia) burst open, germinate on moist surfaces.	*Rhizopus* (Bread Mould).

📌 **Diagram: Binary Fission in Leishmania**



SECTION 2: SEXUAL REPRODUCTION IN FLOWERING PLANTS

Sexual reproduction involves two parents combining specialized germ-cells (gametes) to produce genetically varied offspring.

2.1 Flower Anatomy

- **Stamen (Male Reproductive Part):** Consists of **Anther** (produces pollen grains containing male gametes) and **Filament** (support stalk).
- **Carpel / Pistil (Female Reproductive Part):** Consists of **Stigma** (sticky top receiving pollen), **Style** (elongated middle tube), and **Ovary** (swollen base containing ovules with female egg cells).

📌 Diagram: Longitudinal Section of a Flower

2.2 Pollination & Double Fertilization

- **Pollination:** Transfer of pollen grains from anther to stigma.
  - **Self-pollination:** Transfer within the same flower or plant.
  - **Cross-pollination:** Transfer between different plants (by wind, water, or insects).
- **Fertilization Pathway:** Pollen lands on Stigma → Grows a **Pollen Tube** down through Style → Reaches Ovary → Enters Ovule → Male germ-cell fuses with Female egg cell → Forms **Zygote**.
- **Post-Fertilization Changes:** Zygote becomes **Embryo**, Ovule becomes **Seed** (protected by tough coat), Ovary ripens to become **Fruit**. Petals, sepals, and stamens shrivel and fall off.

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Pollen Grain on Stigma → Germination & Pollen Tube Growth → Enters Ovule → Zygote
Formation → Embryo → Seed & Fruit
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SECTION 3: HUMAN REPRODUCTIVE SYSTEM

Human reproduction involves internal fertilization following sexual maturity reached during **Puberty** (guided by sex hormones: Testosterone in males, Oestrogen in females).

3.1 Male Reproductive System

- **Testes:** Located outside the abdominal cavity in the **Scrotum** because sperm formation requires a temperature 2–3°C lower than normal body temperature. Produces sperms and secretes **Testosterone**.
- **Vas Deferens:** Duct carrying sperms from testes towards the urethra.
- **Prostate Gland & Seminal Vesicles:** Add fluid secretions to nourish sperms and provide a fluid medium for easier transport (forming **Semen**).
- **Urethra:** Common pathway for both urine and semen.

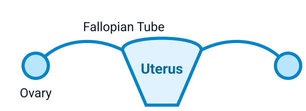
3.2 Female Reproductive System

- **Ovaries:** Pair of organs in lower abdomen producing eggs (ova) and hormones (oestrogen, progesterone). One egg is released every month by one ovary.
- **Oviduct / Fallopian Tube:** Tube where fertilization between sperm and egg takes place.
- **Uterus:** Pear-shaped muscular sac where the fertilized egg embeds (implantation) and develops into a fetus.
- **Placenta:** Disc-like tissue embedded in the uterine wall providing nourishment (O_2, glucose) from mother to embryo and removing metabolic wastes.

3.3 Menstrual Cycle

If the egg is not fertilized, the thickened inner lining of the uterus (endometrium) breaks down and sheds along with blood vessels as **Menstruation**, lasting ~2 to 8 days every month.

📌 **Diagram: Human Female Reproductive System**



SECTION 4: REPRODUCTIVE HEALTH & CONTRACEPTION

4.1 Sexually Transmitted Diseases (STDs)

Causative Pathogen	Disease Examples	Prevention Method
Bacterial Infections	Gonorrhoea, Syphilis	Barrier methods (Condoms) prevent physical fluid exchange.
Viral Infections	Warts, HIV-AIDS	Barrier methods (Condoms), safe medical equipment.

4.2 Contraceptive Methods Comparison Table

Category	Methods / Examples	Working Principle
Barrier Methods	Condoms, Diaphragms, Cervical Caps	Prevents physical contact between sperm and egg cell.
Chemical / Oral Methods	Oral Contraceptive Pills	Alters hormonal balance so eggs are not released; side effects occur.
Intrauterine Devices (IUDs)	Copper-T, Loop	Placed in uterus; prevents implantation by blocking sperm/egg entry.
Surgical Methods	**Vasectomy** (Males: Vas deferens cut) **Tubectomy** (Females: Fallopian tube cut)	Blocks gamete transport permanently; highly effective long-term method.

❓ Important Practice Questions

Q1: Why is DNA copying an essential part of the process of reproduction?
Ans: DNA copying ensures that body designs, cellular organization, and genetic traits are faithfully passed from parent to offspring. Biochemical inaccuracies during copying create essential variations for evolution.

Q2: Why are testes located outside the abdominal cavity in human males?
Ans: Testes are located in the scrotum outside the abdominal cavity because sperm formation requires a lower temperature (2–3°C lower) than normal internal body temperature.

Q3: What are the post-fertilization changes that take place in a flower?
Ans: After fertilization, the zygote forms an embryo inside the ovule. The ovule develops a tough coat to become a seed, and the ovary ripens into a fruit. Sepals, petals, and stamens wither and fall off.

Q4: Differentiate between Vasectomy and Tubectomy.

Ans: Vasectomy is a surgical contraceptive method in males where the vas deferens is cut and ligated. Tubectomy is a surgical method in females where the fallopian tubes are cut and ligated.

Q5: What is the role of Placenta in human embryo development?

Ans: The placenta is a disc-like tissue embedded in the uterine wall. It provides glucose, oxygen, and nutrients from the mother to the developing embryo and carries metabolic wastes from the fetus back into mother's blood.